K-NEAREST NEIGHBOURS (KNN) ALGORITHM

AIM

To understand the KNN algorithm by classifying the species of the Iris flowers.

PROGRAM

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('Iris.csv')
df.info()
print(df.variety.value_counts())
print(df.head())

features = df.iloc[:, :-1].values
label = df.iloc[:, 4].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain, xtest, ytrain, ytest = train_test_split(
    features, label, test_size=.2, random_state=23)

model_KNN = KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain, ytrain)

print("Training score :", model_KNN.score(xtrain, ytrain))
print("Testing score  :", model_KNN.score(xtest, ytest))

from sklearn.metrics import confusion_matrix, classification_report
ypred = model_KNN.predict(xtest)
print("\nConfusion matrix:\n", confusion_matrix(ytest, ypred))
print("\nClassification report:\n", classification_report(ytest, ypred))

# Predicting a new sample
sample = np.array([[5.1, 3.5, 1.4, 0.2]])
print("Prediction for", sample[0], "->", model_KNN.predict(sample)[0])

# Effect of K on accuracy
scores = []
for k in range(1, 16):
    knn = KNeighborsClassifier(n_neighbors=k).fit(xtrain, ytrain)
    scores.append(knn.score(xtest, ytest))
plt.plot(range(1, 16), scores, marker='o')
plt.xlabel('K (number of neighbours)')
plt.ylabel('Testing accuracy')
plt.title('Accuracy for different values of K')
plt.show()
```

OUTPUT

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    str
dtypes: float64(4), str(1)
```

```
memory usage: 6.0 KB
variety
Setosa       50
Versicolor   50
Virginica    50
Name: count, dtype: int64
   sepal.length  sepal.width  petal.length  petal.width variety
0           5.1          3.5           1.4          0.2  Setosa
1           4.9          3.0           1.4          0.2  Setosa
2           4.7          3.2           1.3          0.2  Setosa
3           4.6          3.1           1.5          0.2  Setosa
4           5.0          3.6           1.4          0.2  Setosa
Training score : 0.9583333333333334
Testing score  : 1.0

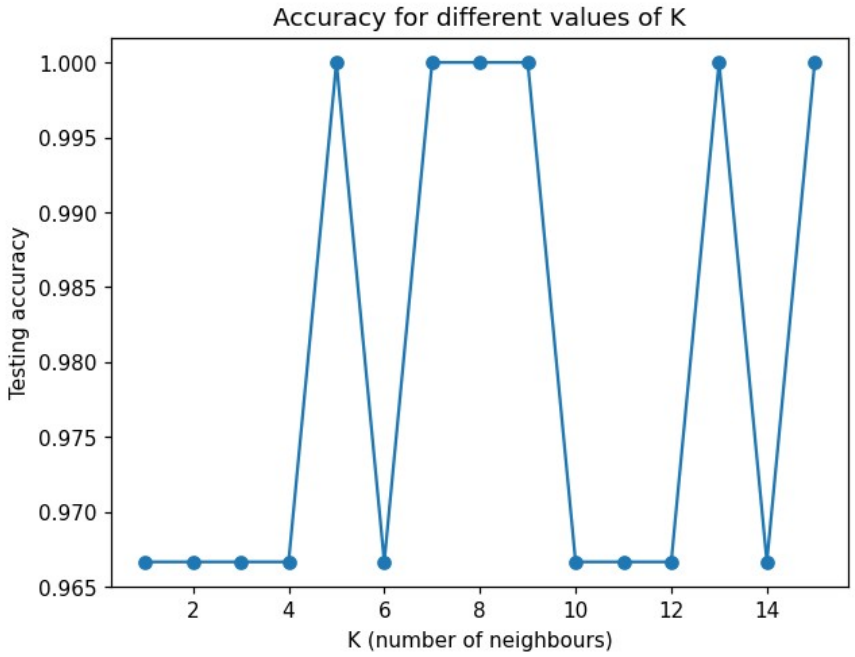
Confusion matrix:
 [[12  0  0]
 [ 0  8  0]
 [ 0  0 10]]

Classification report:
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        12
  Versicolor       1.00      1.00      1.00         8
   Virginica       1.00      1.00      1.00        10

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30

Prediction for [5.1 3.5 1.4 0.2] -> Setosa
```



RESULT

Thus the KNN algorithm was implemented on the Iris data set and the species of the flowers were classified with high accuracy.